

CONDITIONAL STATEMENTS (IF-ELSE)

LEVEL 1: Simple Decisions - Basic If-Else

Program 1: Positive, Negative, or Zero Write a program that takes a number as input and displays whether it is positive, negative, or zero.

Program 2: Even or Odd Write a program that takes a number as input and displays whether it is even or odd.

Program 3: Greater of Two Numbers Write a program that takes two numbers as input and displays which one is greater, or if they are equal.

Program 4: Divisibility Check Write a program that takes two numbers (dividend and divisor) and checks if the dividend is divisible by the divisor. Display "Divisible" or "Not Divisible".

Program 5: Vowel or Consonant Write a program that takes a single character as input and checks if it is a vowel (a, e, i, o, u, A, E, I, O, U) or consonant. Display the result.

Program 6: Alphabet, Digit, or Special Character Write a program that takes a single character as input and checks if it is an alphabet (A-Z, a-z), digit (0-9), or special character. Display the result.

Program 7: Uppercase or Lowercase Write a program that takes a single alphabet character and checks if it is uppercase or lowercase. Display the result.

Program 8: Pass or Fail Write a program that takes a student's marks (out of 100) as input and displays "Pass" if marks ≥ 40 , otherwise displays "Fail".

Program 9: Eligibility for Voting Write a program that takes a person's age as input and checks if they are eligible to vote (age ≥ 18). Display "Eligible" or "Not Eligible".

Program 10: Leap Year Checker Write a program that takes a year as input and checks if it is a leap year. A year is leap if divisible by 4 AND (not divisible by 100 OR divisible by 400). Display "Leap Year" or "Not Leap Year".

Program 11: Triangle Validity by Angles Write a program that takes three angles as input and checks if they can form a valid triangle (sum of angles = 180). Display "Valid Triangle" or "Invalid Triangle".

Program 12: Profit, Loss, or No Profit No Loss Write a program that takes cost price and selling price as input and displays whether there is profit, loss, or no profit no loss.

LEVEL 2: Multiple Conditions & Comparisons

Program 13: Greatest of Three Numbers Write a program that takes three numbers as input and displays the greatest among them.

Program 14: Smallest of Three Numbers Write a program that takes three numbers as input and displays the smallest among them.

Program 15: Check if Number is Single Digit, Double Digit, or More Write a program that takes a number and checks if it is a single digit (0-9), double digit (10-99), or more than double digit. Display the category.

Program 16: Grade Calculator Write a program that takes marks (0-100) as input and displays grade: A (90-100), B (80-89), C (70-79), D (60-69), E (40-59), F (0-39).

Program 17: Triangle Type by Sides Write a program that takes three sides of a triangle as input and checks if it forms an Equilateral (all sides equal), Isosceles (two sides equal), or Scalene (all sides different) triangle.

Program 18: Triangle Validity by Sides Write a program that takes three sides as input and checks if they can form a valid triangle (sum of any two sides > third side). Display "Valid" or "Invalid".

Program 19: Quadrant Finder Write a program that takes x and y coordinates as input and displays which quadrant the point lies in (I, II, III, IV), or if it lies on an axis or origin.

Program 20: BMI Category Write a program that takes BMI value as input and displays category: Underweight (<18.5), Normal (18.5-24.9), Overweight (25-29.9), Obese (≥ 30).

Program 21: Day of Week Identifier Write a program that takes a number (1-7) and displays the corresponding day of the week (1=Monday, 2=Tuesday, ..., 7=Sunday). Display "Invalid" for other numbers.

Program 22: Month Days Calculator Write a program that takes month number (1-12) and year as input, and displays the number of days in that month (handle leap year for February).

Program 23: Character Position in Alphabet (First or Second Half) Write a program that takes an alphabet character (A-Z or a-z) and displays if it falls in first half (A-M) or second half (N-Z) of alphabets.

Program 24: Divisibility by 2, 3, 5, or None Write a program that takes a number and checks if it is divisible by 2, 3, 5, or none of these. Display appropriate message for each case.

Program 25: Number Range Classifier Write a program that takes a number and classifies it into ranges: 1-10, 11-50, 51-100, 101-500, 501-1000, or above 1000. Display the range.

Program 26: Valid Date Checker (Simple) Write a program that takes day (1-31) and month (1-12) and checks basic validity: day should be 1-31 for most months, 1-30 for Apr/Jun/Sep/Nov, 1-28/29 for Feb. Display "Valid" or "Invalid".

Program 27: Electricity Bill with Slab Rates Write a program that calculates electricity bill based on units: 0-100 units: ₹5/unit, 101-200: ₹7/unit, 201-300: ₹10/unit, above 300: ₹15/unit. Display total bill.

Program 28: Check if Number is Divisible by Both 3 and 5 Write a program that takes a number and checks: if divisible by both 3 and 5 display "Divisible by Both", if by 3 only "Divisible by 3", if by 5 only "Divisible by 5", else "Not Divisible by 3 or 5".

LEVEL 3: Nested Conditions & Complex Logic

Program 29: Triangle Type by Angles Write a program that takes three angles of a triangle and displays whether it is Acute (all angles < 90°), Right (one angle = 90°), or Obtuse (one angle > 90°) triangle.

Program 30: Ascending or Descending Order Checker Write a program that takes three numbers and checks if they are in strictly ascending order, strictly descending order, or neither.

Program 31: Roots of Quadratic Equation Type Write a program that takes coefficients a , b , c of quadratic equation $ax^2+bx+c=0$, calculates discriminant (b^2-4ac), and displays whether roots are real & distinct ($D>0$), real & equal ($D=0$), or imaginary ($D<0$).

Program 32: Income Tax Calculator (Simple Slabs) Write a program that calculates tax based on income: 0-2.5L: 0%, 2.5L-5L: 5%, 5L-10L: 20%, above 10L: 30%. Display tax amount.

Program 33: Ticket Pricing with Age-Based Discounts Write a program that calculates ticket price based on age: Child (0-12): ₹100, Teen (13-17): ₹150, Adult (18-59): ₹200, Senior (60+): ₹120. Display the ticket price.

Program 34: Salary Increment Calculator Write a program that calculates salary increment based on performance rating (1-5): Rating 5: 20%, Rating 4: 15%, Rating 3: 10%, Rating 2: 5%, Rating 1: 0%. If years of service > 5, add extra 5%. Display new salary.

Program 35: Loan Eligibility Checker Write a program that checks loan eligibility based on: age (21-60), income (>25000), credit score (>700). Display "Eligible" only if all conditions are met, otherwise display which condition failed.

Program 36: Strong Password Validator (Simple) Write a program that takes password length, count of uppercase letters, lowercase letters, and digits as separate inputs. Check if password is Strong (length $\geq$ 8, has uppercase, lowercase, digits), Medium (length $\geq$ 6, any 2 types), or Weak.

Program 37: Admission Eligibility (Multiple Criteria) Write a program that checks college admission eligibility based on: Maths marks $\geq$ 80, Physics marks $\geq$ 75, Chemistry marks $\geq$ 75, Total of 3 subjects $\geq$ 240. Display "Eligible" if all conditions met, else "Not Eligible".

Program 38: Discount Calculator with Multiple Conditions Write a program that calculates discount based on: Purchase amount > ₹10000: 20% discount, > ₹5000: 15%, > ₹2000: 10%, else no discount. If customer is a member, add 5% extra discount. Display final price after discount.

Program 39: Shipping Cost Calculator Write a program that calculates shipping cost based on: weight (kg) and distance (km). Base cost: ₹50. Add ₹10/kg if weight > 5kg. Add ₹5/km if distance > 100km. If both conditions true, add ₹20 extra. Display total shipping cost.

Program 40: Water Bill with Conservation Incentive Write a program that calculates water bill: 0-5000L: ₹2/L, 5001-10000L: ₹3/L, above 10000L: ₹5/L. If usage < 3000L, give 15% discount (conservation reward). Display total bill.